Teaching Approach Analysis

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Load data:

survey_data

- survey responses from qualtrics
- including Question 4d) How would you describe the dominant approach to teaching Data Analysis?

```
survey_data <- read_rds(here("data/clean_data", "survey_data.rds"))

#look at data
survey_data %>%
  select(university_id, discipline, teaching_approach) %>%
  glimpse()
```

```
## Rows: 103
## Columns: 3
## $ university_id     <fct> u90, u90, u67, u67, u66, u67, u97, u86, u93, u12, u8~
## $ discipline        <fct> Bio, Psych, Bio, Med, Bio, Psych, Biomed, Biomed, Me~
## $ teaching_approach <ord> allFunc, emphMath, allFunc, allFunc, allFunc, emphMa~
```

Prepare for model

```
#select columns
survey_filt <- survey_data %>%
  select(university_id, discipline, teaching_approach)

#summary
summary(survey_filt)
```

```
##  university_id  discipline     teaching_approach
##  u67    : 4    Bio   :31   allMath      : 1
##  u103   : 3    Biomed:19   emphMath     : 5
##  u37    : 3    Med   :15   equalMathFunc:20
##  u58    : 3    Psych :38   emphFunc     :29
##  u59    : 3                allFunc      :48
##  (Other):82
##  NA's   : 5
```

```
#structure: ordered factor
str(survey_filt$teaching_approach)
```

```
##  Ord.factor w/ 5 levels "allMath"<"emphMath"<..: 5 2 5 5 5 2 4 3 4 4 ...
```

```
#table: discipline by teaching approach
table(survey_filt$discipline, survey_filt$teaching_approach)
```

```
##
##          allMath emphMath equalMathFunc emphFunc allFunc
##   Bio          0        1             6        5      19
##   Biomed       1        0             6        8       4
##   Med          0        0             2        6       7
##   Psych        0        4             6       10      18
```

Note:

- `allMath` means "All Math"
- `emphMath` means "Emphasizes Math"
- `equalMathFunc` means "Equal Math and Functional"
- `emphFunc` means "Emphasizes Functional"
- `allFunc` means "All Functional"

The original 5 categories

The survey provided the opportunity to classify the teaching approach in 5 ways: Mathematical (all Math), a mix of mathematical and functional perspectives that emphasizes math (emphMath), an equal balance of mathematical and functional approaches (equalMathFunct), a mix of functional and mathematical perspectives that emphasized functional (emphFunct), or a functional approach (allFunct). The table, above, documents the raw data from this survey.

In order to reduce issues that arise from categories with zero responses, (e.g. no medical school reported a purely mathematical teaching approach), we re-code teaching approach into 3 categories: Mathematical; equalMathFunc; Functional.

```
#recode into 3 categories
survey_filt <- survey_filt %>%
  mutate(teaching_approach_3 =
           fct_recode(teaching_approach,
            "Mathematical" =  "allMath",
            "Mathematical" = "emphMath",
            "Equal" = "equalMathFunc",
            "Functional" = "emphFunc",
            "Functional" = "allFunc"
            )
         )

#table of teaching approach (3 categories) by discipline
table(survey_filt$discipline, survey_filt$teaching_approach_3)
```

```
##
##          Mathematical Equal Functional
##   Bio               1     6         24
##   Biomed            1     6         12
##   Med               0     2         13
##   Psych             4     6         28
```

Multinomial model of 3 categories

We now fit a multinomial model, using a prior recommended by Gelman et al. (2008) (DOI: 10.1214/08-AOAS191).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```r
#We obtain the default prior, and then update the prior, below.
get_prior(teaching_approach_3 ~ discipline, family = categorical(link = "logit"), data = survey_filt)
```

```
##                 prior     class             coef group resp         dpar nlpar
##                (flat)         b                                    muEqual
##                (flat)         b disciplineBiomed                   muEqual
##                (flat)         b    disciplineMed                   muEqual
##                (flat)         b  disciplinePsych                   muEqual
##  student_t(3, 0, 2.5) Intercept                                    muEqual
##                (flat)         b                                muFunctional
##                (flat)         b disciplineBiomed               muFunctional
##                (flat)         b    disciplineMed               muFunctional
##                (flat)         b  disciplinePsych               muFunctional
##  student_t(3, 0, 2.5) Intercept                                muFunctional
##  lb ub tag       source
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
```

```r
#Our prior follows Gelman et al. (2008) (DOI: 10.1214/08-AOAS191):
priors <- c(
  prior(cauchy(0, 10), class = "Intercept", dpar = "muEqual"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "muFunctional"),
  prior(cauchy(0, 2.5), class = "b", dpar = "muEqual"),
  prior(cauchy(0, 2.5), class = "b", dpar = "muFunctional")
)


contrasts(survey_filt$discipline) =contr.sum(4)
contrasts(survey_filt$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

```r
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1

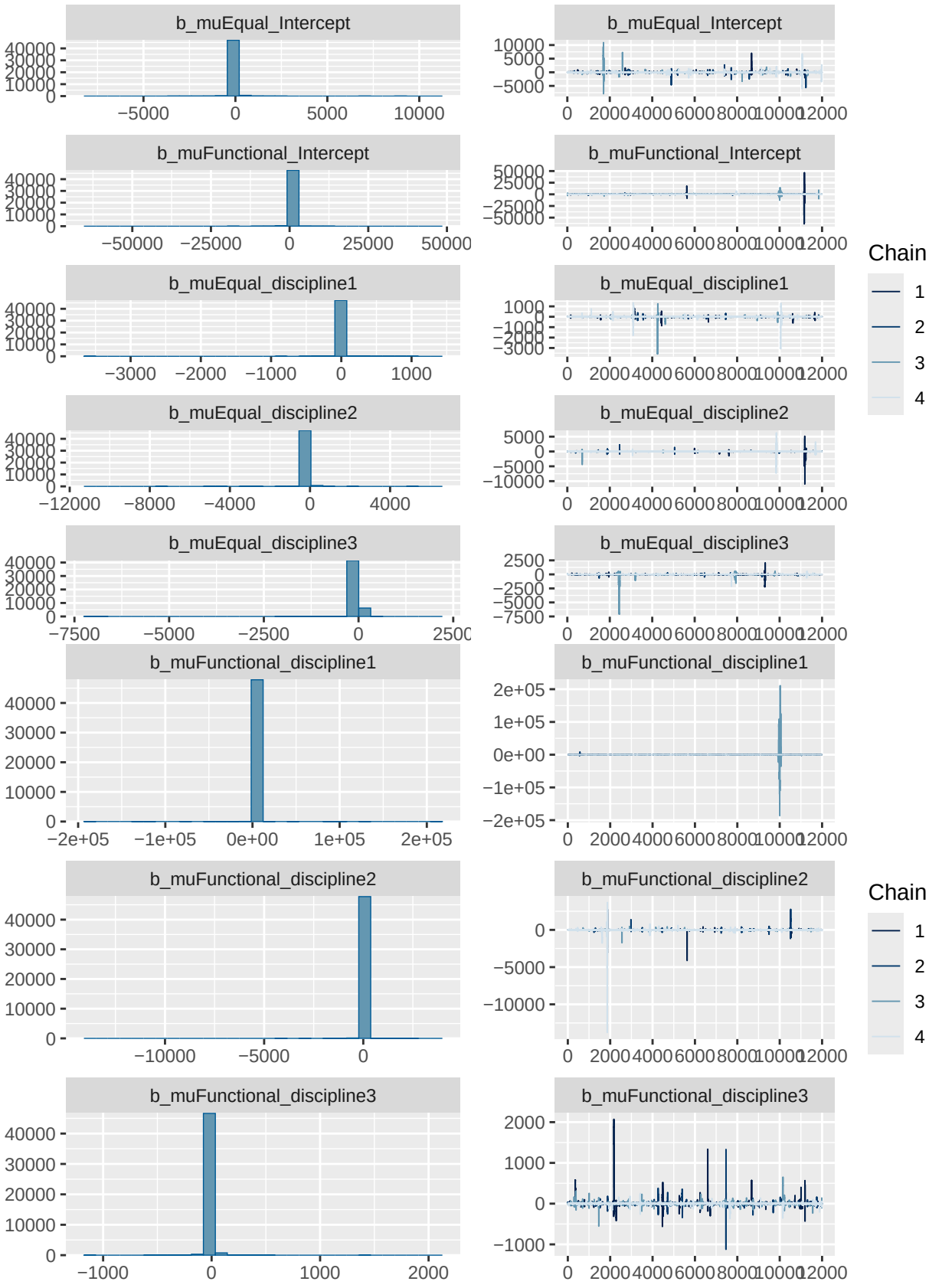

#Prior predictive check:
brm.formula1 <- brmsformula(teaching_approach_3 ~ discipline,
                    family = categorical(link = "logit"))
```

```
brm.gelman.priorpc <- brm(brm.formula1, prior = priors, sample_prior = "only",
                          iter = 24000,
                      control = list(adapt_delta = 0.999, stepsize = 0.001,
                                     max_treedepth = 20),
                      data = survey_filt,
                      file = here("models.gelman", "brm.mn.approach.priorpc"))

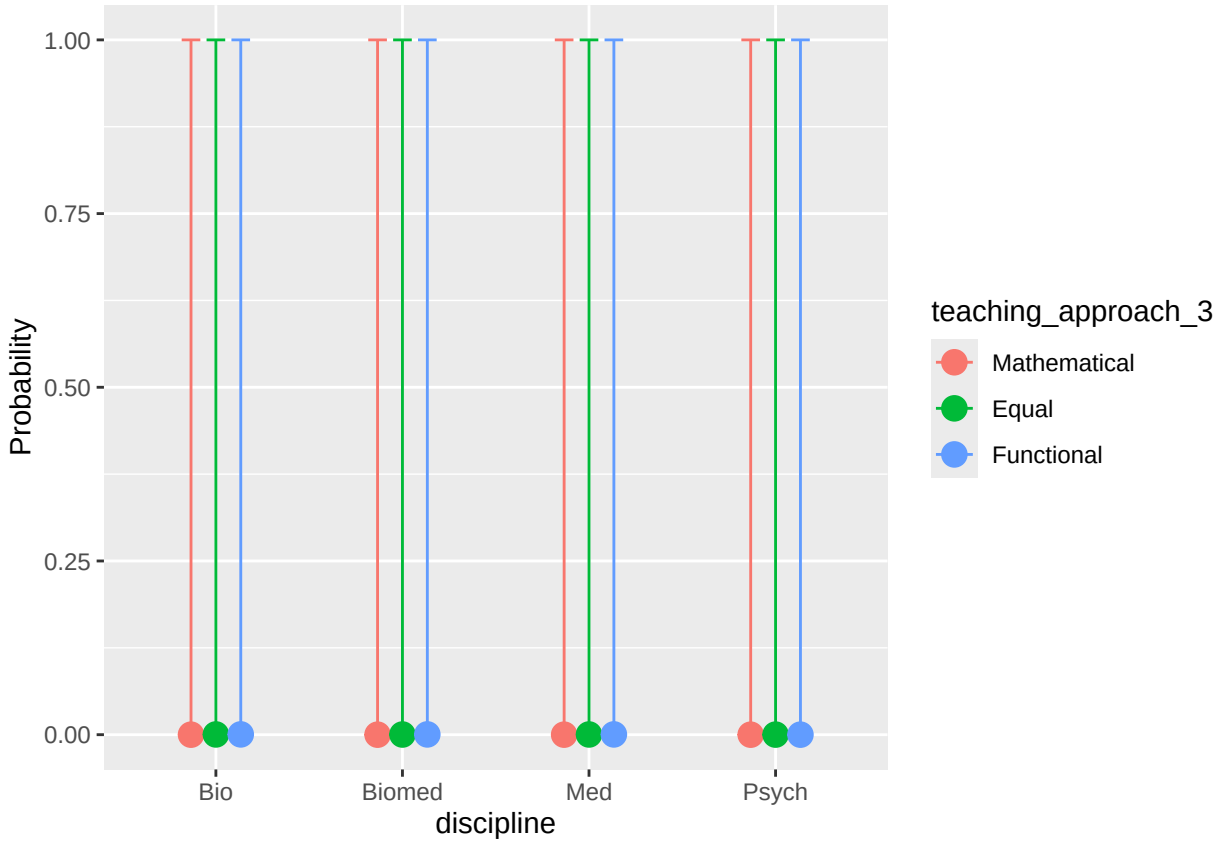
summary(brm.gelman.priorpc)
```

```
##  Family: categorical
##   Links: muEqual = logit; muFunctional = logit
## Formula: teaching_approach_3 ~ discipline
##    Data: survey_filt (Number of observations: 103)
##   Draws: 4 chains, each with iter = 24000; warmup = 12000; thin = 1;
##          total post-warmup draws = 48000
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muEqual_Intercept             1.20    279.81  -153.54   132.15 1.00    16696
## muFunctional_Intercept      -21.53   1334.87  -170.65   145.20 1.00    13328
## muEqual_discipline1          -1.65     91.48   -35.40    29.92 1.00    17017
## muEqual_discipline2          -6.35    245.51   -44.28    32.16 1.00    17360
## muEqual_discipline3          -7.35    190.05   -36.89    30.99 1.00    11378
## muFunctional_discipline1      3.35   5031.31   -30.62    27.74 1.00    16594
## muFunctional_discipline2     -2.08    137.33   -33.76    35.11 1.00    19006
## muFunctional_discipline3      0.33     40.92   -30.44    27.21 1.00    18620
##                           Tail_ESS
## muEqual_Intercept             4354
## muFunctional_Intercept        3051
## muEqual_discipline1           4388
## muEqual_discipline2           4117
## muEqual_discipline3           2740
## muFunctional_discipline1      4045
## muFunctional_discipline2      5647
## muFunctional_discipline3      5226
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(brm.gelman.priorpc)
```


b_muEqual_Intercept
b_muFunctional_Intercept
b_muEqual_discipline1
b_muEqual_discipline2
b_muEqual_discipline3
b_muFunctional_discipline1
b_muFunctional_discipline2
b_muFunctional_discipline3
Chain
1
2
3
4

```
#Prior predictive check outout:
conditional_effects(brm.gelman.priorpc, categorical = TRUE)
```



Output from `summary(brm.gelman.priorpc)` and `plot(brm.gelman.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

NOTE: exploratory analyses (not shown) indicate that stronger priors yield estimates that approach 33% for each discipline (with narrower credible intervals) in prior predictive checks (`conditional_effects(brm.gelman.priorpc, categorical = TRUE)`).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman prior:
brm.mn.approach <- brm(teaching_approach_3 ~ discipline,
                       family = categorical(link = "logit"),
                       prior = priors,
                       iter = 6000,
                       control = list(adapt_delta = 0.999, stepsize = 0.001,
                                      max_treedepth = 20),
                       data = survey_filt,
                       file = here("models.gelman", "brm.mn.approach"))

#Examine prior that we specified:
prior_summary(brm.mn.approach)
```

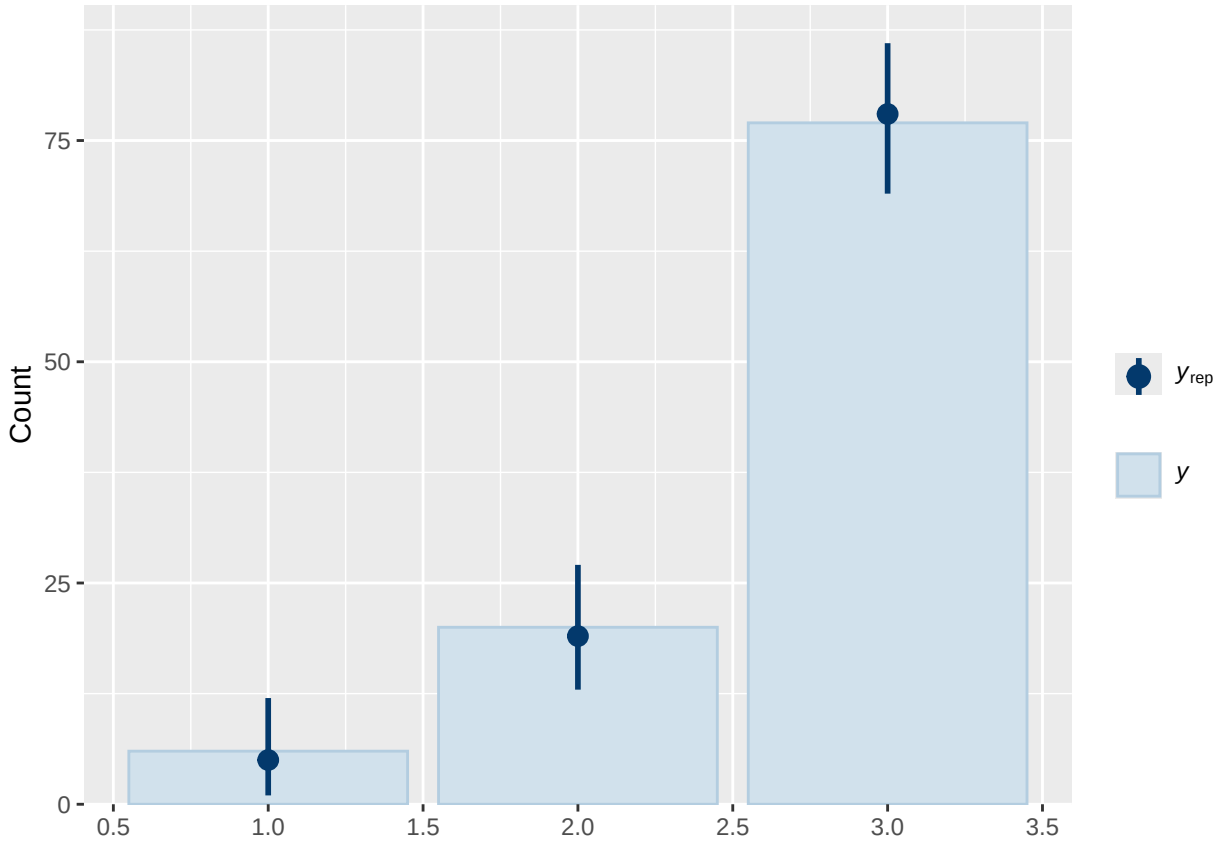
```
##           prior     class        coef group resp         dpar nlpar lb ub tag
##  cauchy(0, 2.5)         b                             muEqual
##  cauchy(0, 2.5)         b disciplined1                muEqual
##  cauchy(0, 2.5)         b disciplined2                muEqual
```

```
##  cauchy(0, 2.5)         b discipline3                muEqual
##  cauchy(0, 2.5)         b                         muFunctional
##  cauchy(0, 2.5)         b discipline1             muFunctional
##  cauchy(0, 2.5)         b discipline2             muFunctional
##  cauchy(0, 2.5)         b discipline3             muFunctional
##   cauchy(0, 10) Intercept                             muEqual
##   cauchy(0, 10) Intercept                          muFunctional
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
```

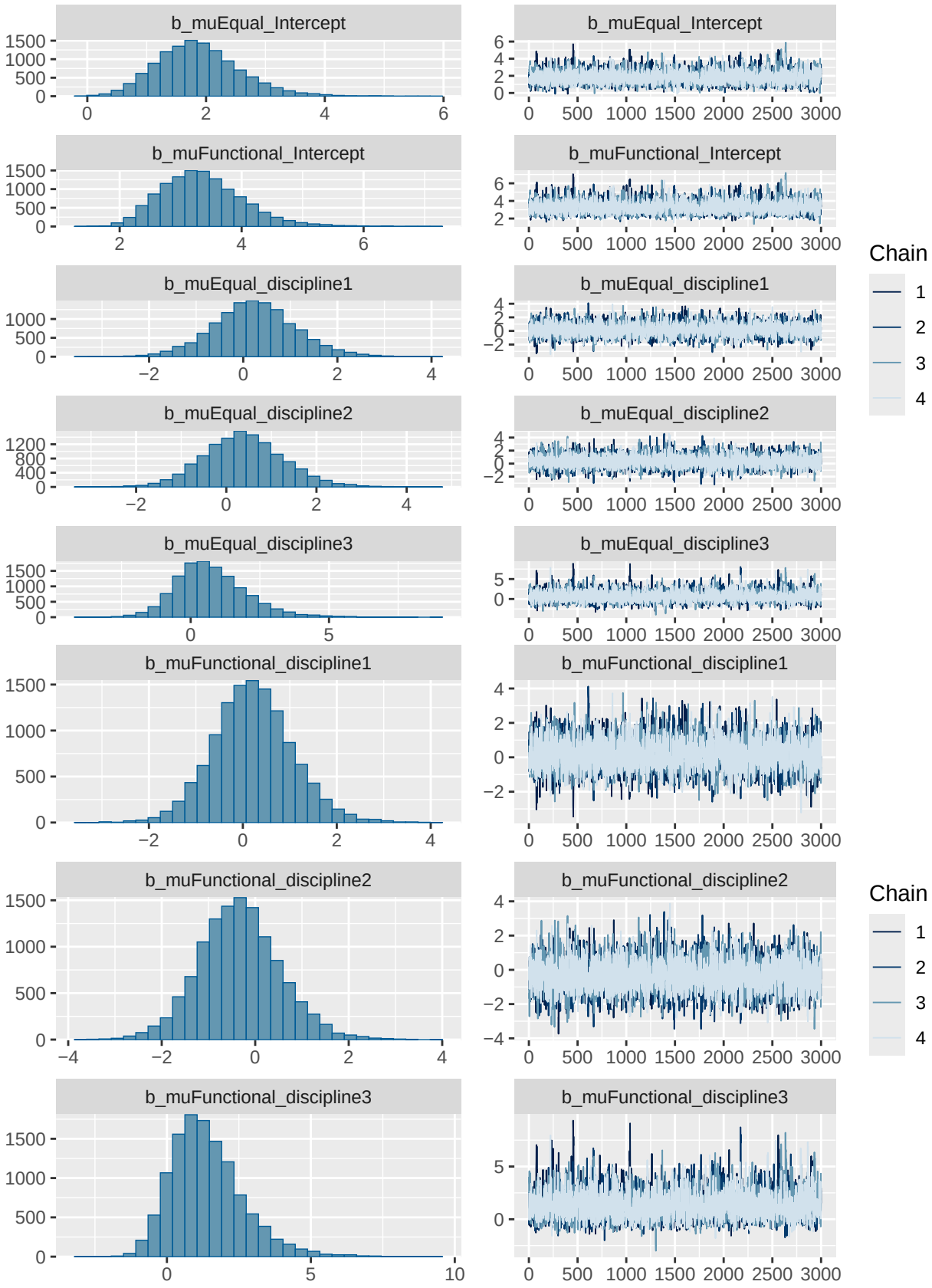
```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.approach)
```

```
##  Family: categorical
##   Links: muEqual = logit; muFunctional = logit
## Formula: teaching_approach_3 ~ discipline
##    Data: survey_filt (Number of observations: 103)
##   Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muEqual_Intercept            1.88      0.71     0.65     3.44 1.00     3277
## muFunctional_Intercept       3.36      0.68     2.23     4.89 1.00     3189
## muEqual_discipline1          0.25      0.89    -1.49     2.09 1.00     3507
## muEqual_discipline2          0.39      0.90    -1.32     2.24 1.00     4473
## muEqual_discipline3          0.76      1.31    -1.44     3.70 1.00     3274
## muFunctional_discipline1     0.22      0.85    -1.42     1.97 1.00     3530
## muFunctional_discipline2    -0.34      0.87    -2.01     1.45 1.00     4223
## muFunctional_discipline3     1.40      1.27    -0.61     4.37 1.00     3199
##                          Tail_ESS
## muEqual_Intercept            3331
## muFunctional_Intercept       3317
## muEqual_discipline1          4414
## muEqual_discipline2          4411
## muEqual_discipline3          3005
## muFunctional_discipline1     4217
## muFunctional_discipline2     4443
## muFunctional_discipline3     2964
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Posterior predictive check:
pp_check(brm.mn.approach, type = "bars", ndraws = 100)
```




```r
#Check traces
plot(brm.mn.approach)
```



The summary (`summary(brm.mn.approach)`) indicates that the model converged well (based on Rhat) and

has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.approach, type = "bars", ndraws = 100)`) indicates sufficient model fit.

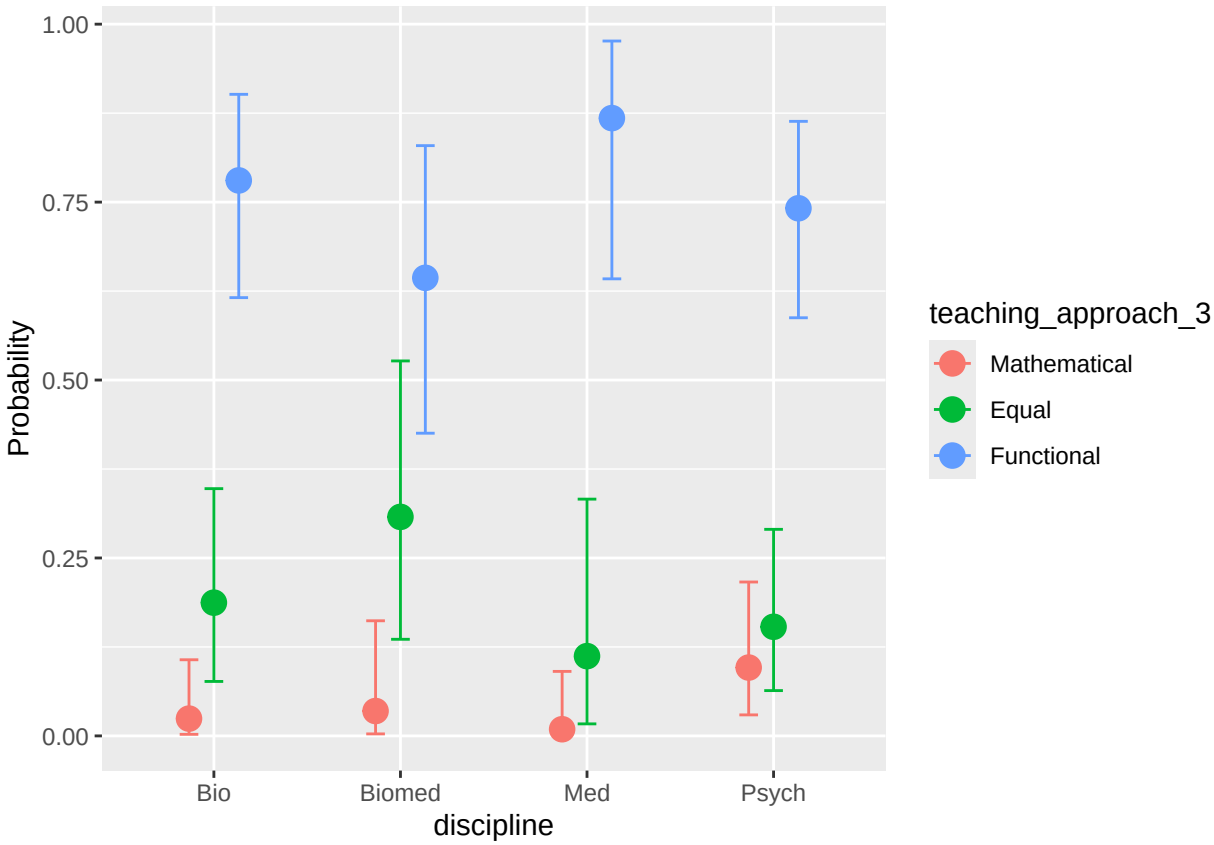
The traces and distributions of estimates on the latent scale (`plot(brm.mn.approach)`) also indicate good model behaviour.

Now we obtain the model estimates:

```
eff_brm.mn.approach <- conditional_effects(brm.mn.approach, categorical = TRUE)
eff_brm.mn.approach[[1]]
```

```
##    discipline teaching_approach_3 cond__       cats__ effect1__    effect2__
## 1         Bio        Mathematical      1 Mathematical       Bio Mathematical
## 2      Biomed        Mathematical      1 Mathematical    Biomed Mathematical
## 3         Med        Mathematical      1 Mathematical       Med Mathematical
## 4       Psych        Mathematical      1 Mathematical     Psych Mathematical
## 5         Bio        Mathematical      1        Equal       Bio         Equal
## 6      Biomed        Mathematical      1        Equal    Biomed         Equal
## 7         Med        Mathematical      1        Equal       Med         Equal
## 8       Psych        Mathematical      1        Equal     Psych         Equal
## 9         Bio        Mathematical      1   Functional       Bio   Functional
## 10     Biomed        Mathematical      1   Functional    Biomed   Functional
## 11        Med        Mathematical      1   Functional       Med   Functional
## 12      Psych        Mathematical      1   Functional     Psych   Functional
##     estimate__       se__      lower__    upper__
## 1  0.024313455 0.02113905 0.0022213968 0.10697017
## 2  0.034976776 0.03154882 0.0028127682 0.16175988
## 3  0.009385561 0.01159402 0.0001237096 0.09064909
## 4  0.096035988 0.04681020 0.0295254364 0.21625404
## 5  0.187218670 0.06936666 0.0765191218 0.34739841
## 6  0.307726400 0.10522106 0.1357685368 0.52685639
## 7  0.112094128 0.07662103 0.0169242271 0.33270329
## 8  0.153330610 0.05746503 0.0636876761 0.29022346
## 9  0.780468042 0.07336465 0.6158025680 0.90142477
## 10 0.643612574 0.10953523 0.4253002476 0.82932980
## 11 0.868037257 0.08347243 0.6421981080 0.97634121
## 12 0.741336636 0.07237852 0.5875155720 0.86348012
```

```
print(eff_brm.mn.approach)
```



Plot

```
plot_colour <- "skyblue4"

plot_theme <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.x = element_text(size = 18),
    axis.text.x = element_text(size= 14),
    axis.title.y = element_text(size= 24),
    axis.text.y = element_markdown(size = 16, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 20),
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")
```

```
approach_estimates <- eff_brm.mn.approach[[1]] %>%
  clean_names() %>%
  rename(
    teaching_app_3 = cats,
    probability_estimate = estimate) %>%
  select(discipline, teaching_app_3, probability_estimate, lower, upper)
```

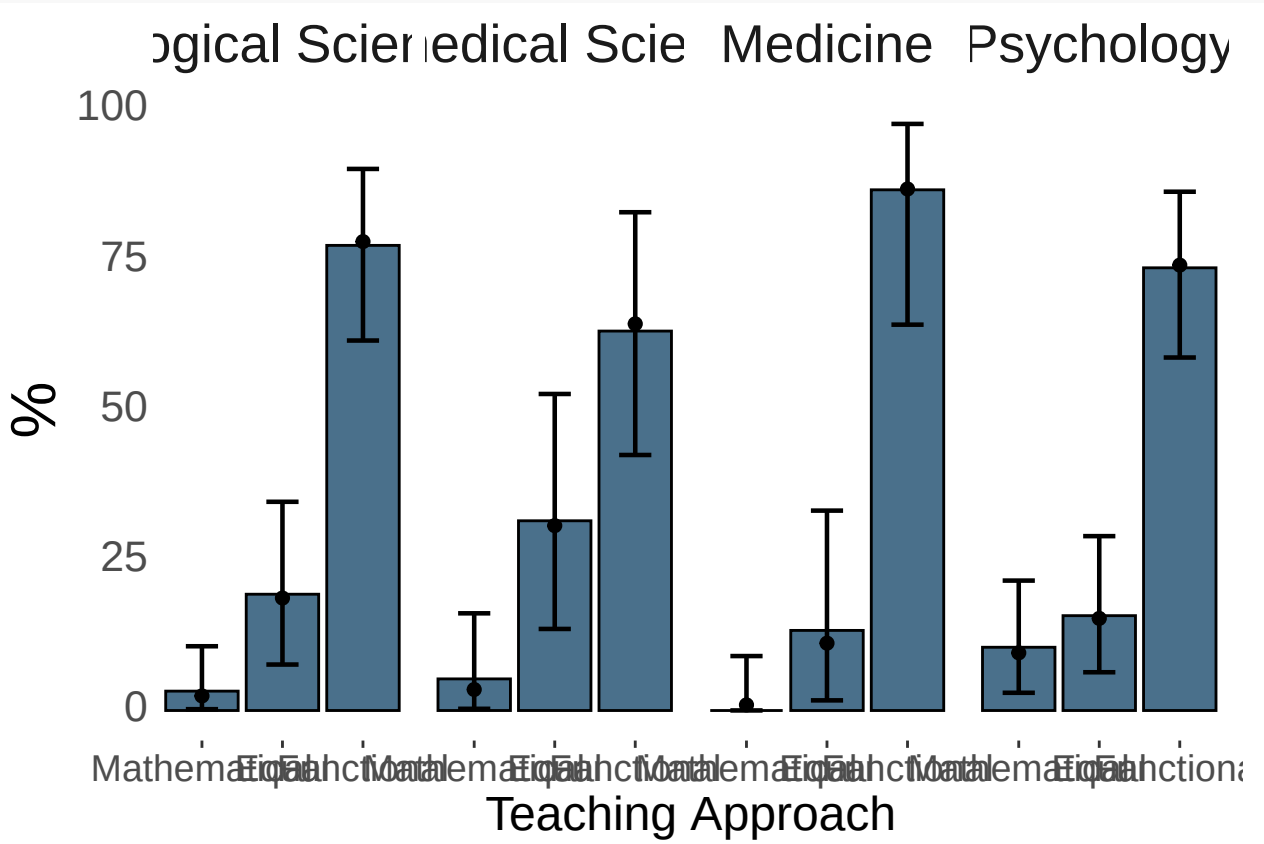
```
approach_estimates
```

```
##    discipline teaching_app_3 probability_estimate        lower      upper
## 1         Bio   Mathematical          0.024313455 0.0022213968 0.10697017
## 2      Biomed   Mathematical          0.034976776 0.0028127682 0.16175988
## 3         Med   Mathematical          0.009385561 0.0001237096 0.09064909
## 4       Psych   Mathematical          0.096035988 0.0295254364 0.21625404
## 5         Bio          Equal          0.187218670 0.0765191218 0.34739841
## 6      Biomed          Equal          0.307726400 0.1357685368 0.52685639
## 7         Med          Equal          0.112094128 0.0169242271 0.33270329
## 8       Psych          Equal          0.153330610 0.0636876761 0.29022346
## 9         Bio     Functional          0.780468042 0.6158025680 0.90142477
## 10     Biomed     Functional          0.643612574 0.4253002476 0.82932980
## 11        Med     Functional          0.868037257 0.6421981080 0.97634121
## 12      Psych     Functional          0.741336636 0.5875155720 0.86348012
```

```
survey_table_app <- table(survey_filt$teaching_approach_3, survey_filt$discipline)
survey_table_app <- prop.table(survey_table_app, margin = 2)
survey_table_app <- as.data.frame(survey_table_app)
survey_ggplot_app <- survey_table_app %>%
  rename(
    teaching_app_3 = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
approach_plot <- ggplot() +
  geom_col(
    data = survey_ggplot_app,
    aes(x = teaching_app_3, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = approach_estimates,
    aes(x = teaching_app_3, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = approach_estimates,
    aes(x = teaching_app_3, y = 100*probability_estimate),
    size = 2) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  xlab("Teaching Approach") +
  plot_theme
```

```
approach_plot
```




```
ggsave(
  plot = approach_plot,
  here("figures", "approach_plot_gelman.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)

brm.mn_fixed <- tidy(brm.mn.approach, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %>%
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.approach, effects = "fixed", conf.level = 0.95,
## : some parameter names contain underscores: term naming may be unreliable!
```

```
brm.mn_fixed
```

```
## # A tibble: 8 x 8
##   effect component term           estimate std.error conf.low conf.high ci_width
##   <chr>  <chr>     <chr>             <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
```

```
## 1 fixed  cond      muEqual_Inter~    1.88      0.711    0.646      3.44      2.79
## 2 fixed  cond      muFunctional_~    3.36      0.682    2.23       4.89      2.66
## 3 fixed  cond      muEqual_disci~    0.246     0.893   -1.49       2.09      3.58
## 4 fixed  cond      muEqual_disci~    0.393     0.901   -1.32       2.24      3.55
## 5 fixed  cond      muEqual_disci~    0.758     1.31    -1.44       3.70      5.13
## 6 fixed  cond      muFunctional_~    0.215     0.852   -1.42       1.97      3.38
## 7 fixed  cond      muFunctional_~   -0.336     0.872   -2.01       1.45      3.46
## 8 fixed  cond      muFunctional_~    1.40      1.27    -0.607      4.37      4.98
```

```r
#We have 8 fixed effects; the first 2 are intercepts, and the last 6 are coefficients (b)
#Also, the 2 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 6 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn_fixed %>%
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  ) %>%
  ggplot(aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

